

Historical redlining is associated with increasing geographical disparities in bird biodiversity sampling in the United States

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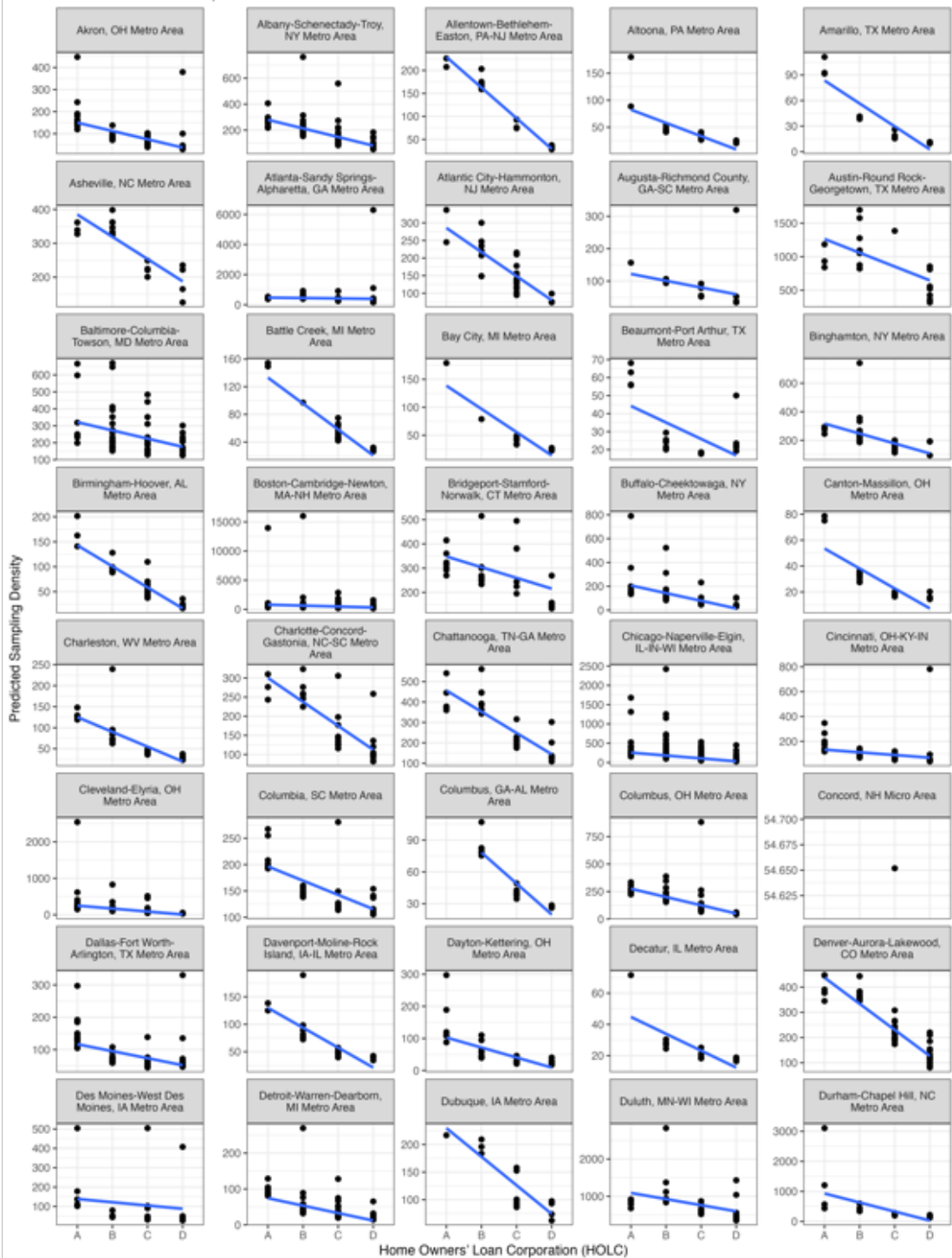
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Predicted Sampling Density of GBIF Observations by HOLC Grade
batch 1: MSAs 1 - 40 in alpha-order

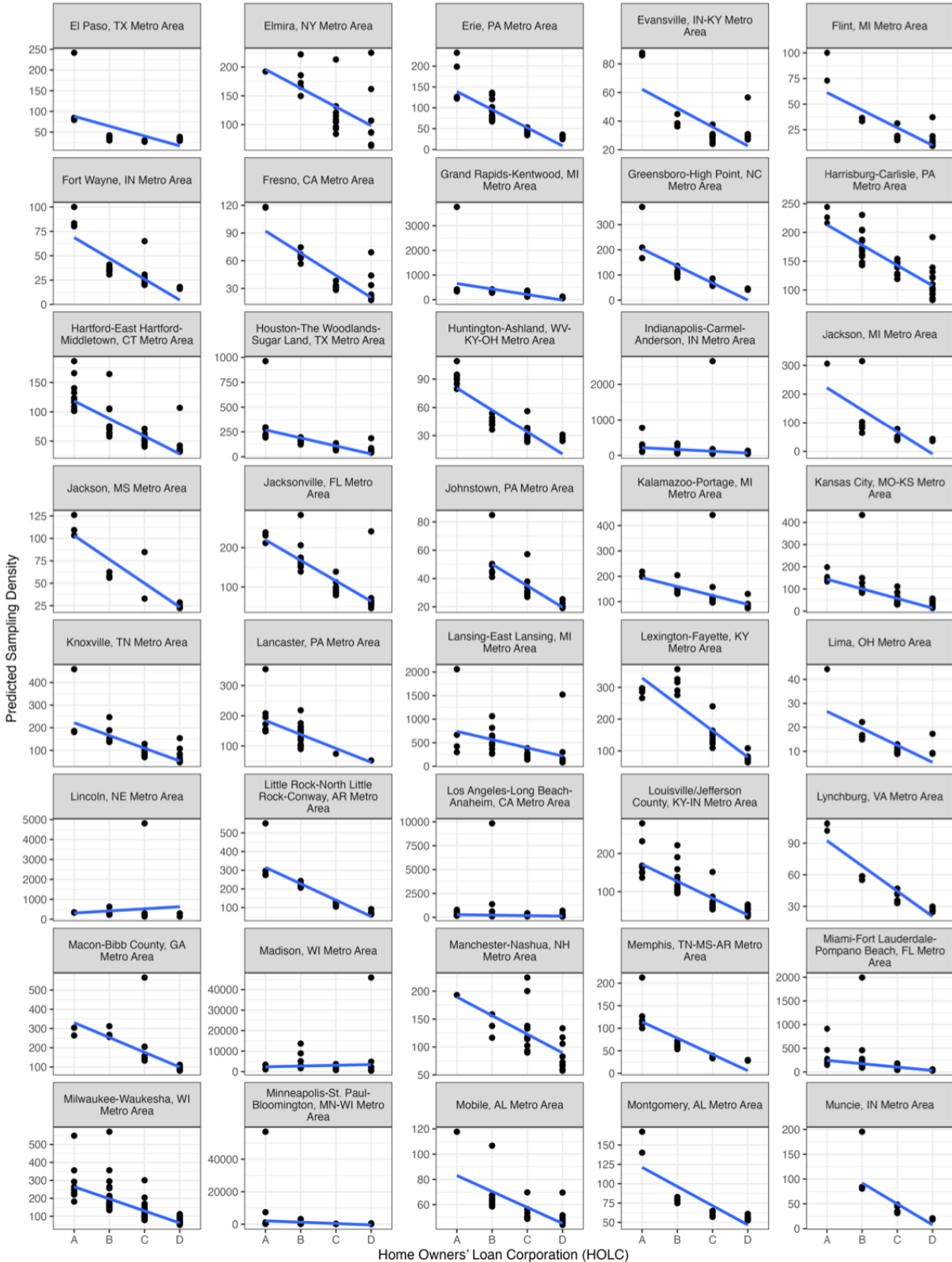


Home Owners' Loan Corporation (HOLC)

model accounts for NDVI at 0.34, % protected area at 3.02%, population per km² at 3.103

Predicted Sampling Density of GBIF Observations by HOLC Grade

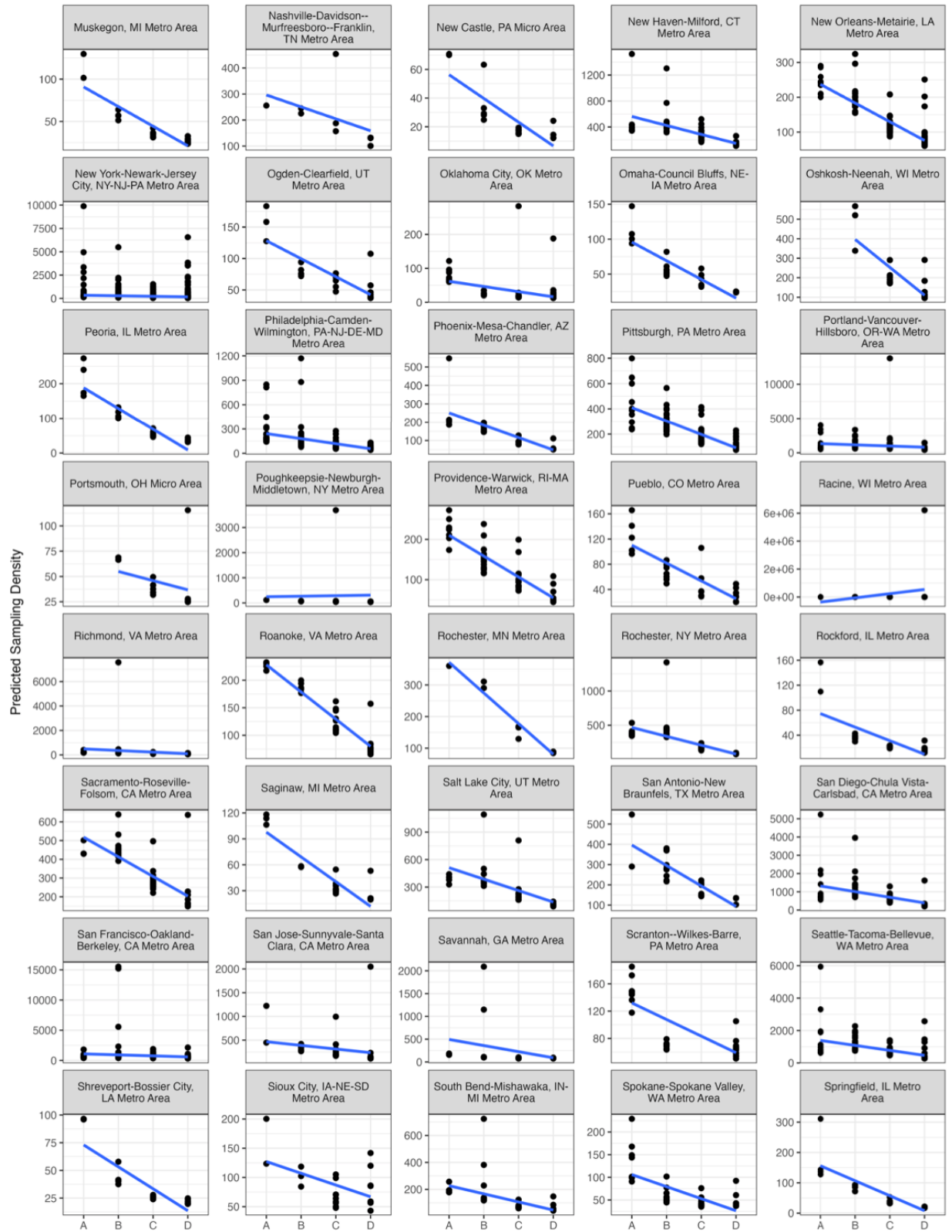
batch 2: MSAs 41 - 80 in alpha-order



model accounts for NDVI at 0.34, % protectect area at 3.02%, population per km² at 3,103

Predicted Sampling Density of GBIF Observations by HOLC Grade

batch 3: MSAs 81 - 120 in alpha-order

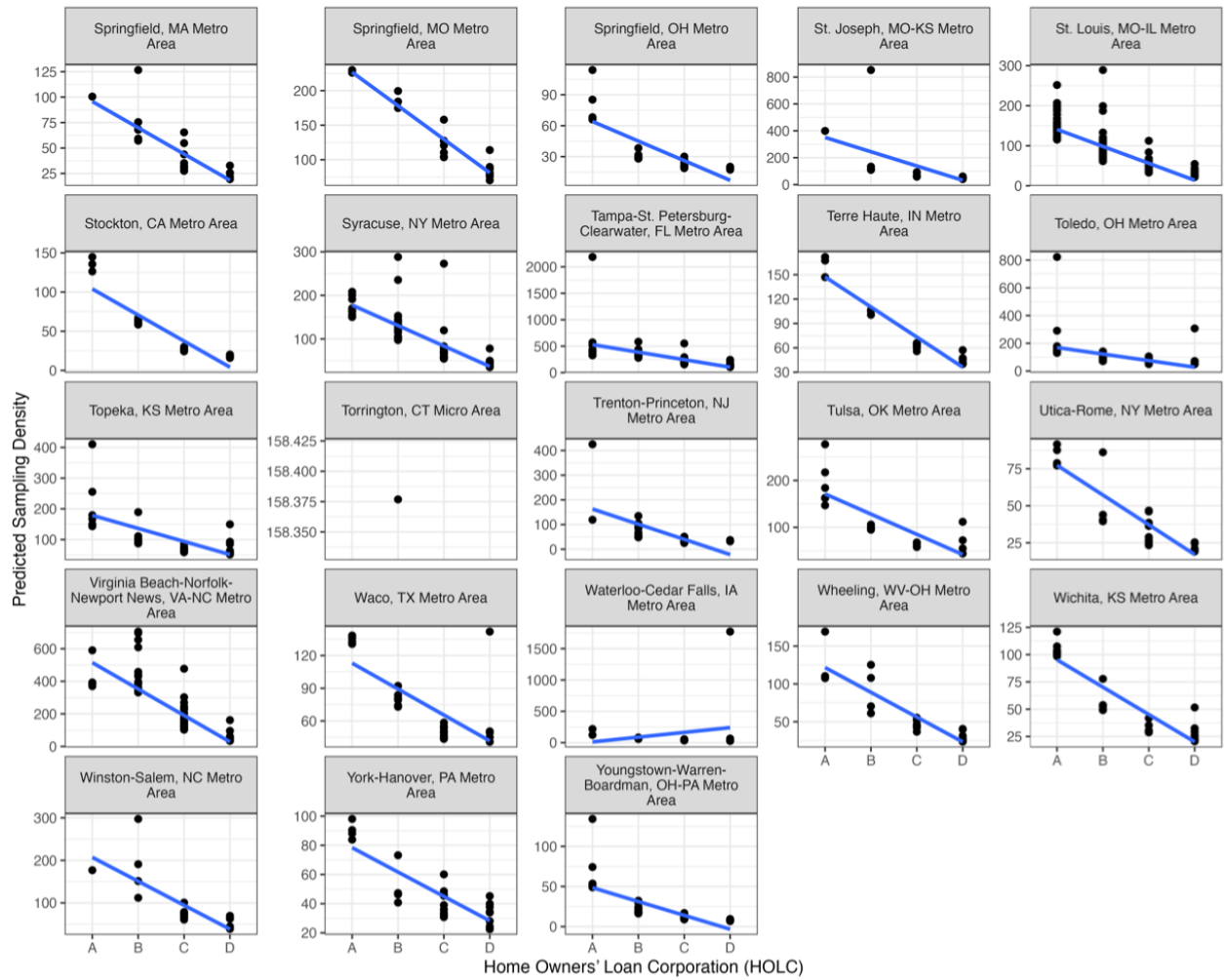


Home Owners' Loan Corporation (HOLC)

model accounts for NDVI at 0.34, % protect area at 3.02%, population per km² at 3,103

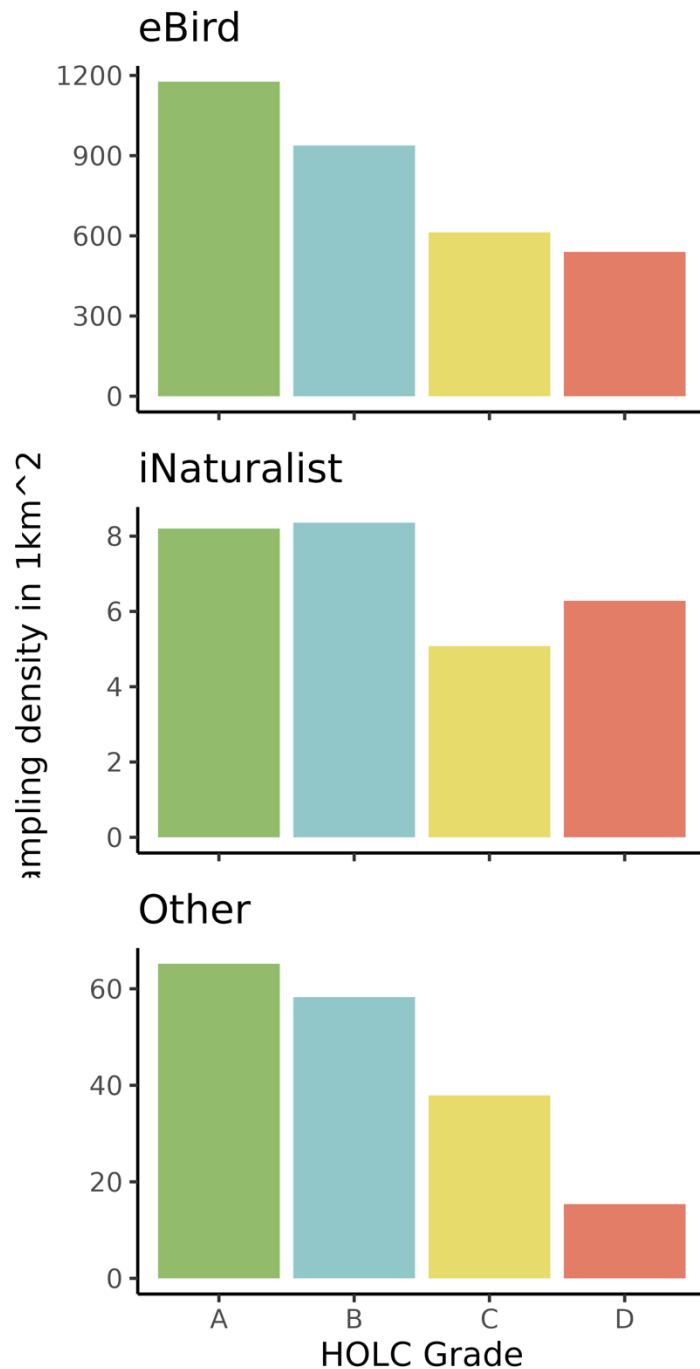
Predicted Sampling Density of GBIF Observations by HOLC Grade

batch 4: MSAs 121 - 143 in alpha-order



model accounts for NDVI at 0.34, % protectect area at 3.02%, population per km^2 at 3,103

Supplemental Figure 1: Predicted sampling density of bird observation per km² by Home Owners' Loan Corporation (HOLC) grades across 196 Metropolitan Statistical Areas (MSA).



Supplementary Figure 2: Patterns of bird biodiversity across redlined areas by observation type. Sampling density (bird records per km²) differs substantially across biodiversity collection type using bird data from 2000-2020. The citizen science platform eBird (top) contributes by far most of the bird biodiversity data across studied cities by orders of magnitude and sampling is highest in HOLC A and lowest in HOLC D neighborhood. The iNaturalist (middle) platform has higher sampling density in HOLC A and B neighborhood than HOLC C and D, but differences are substantially lower. Bird biodiversity sampling density of type 'Other' (bottom) follows the same

trend as eBird, with highest sampling in HOLC A and B and lowest sampling in HOLC C and D neighborhoods.

Supplementary Table 1: Summary statistics of predictor variables employed for mixed effect modeling.

Characteristic	Overall, N = 9,847¹	A, N = 1,123¹	B, N = 2,493¹	C, N = 3,757¹	D, N = 2,474¹
NDVI	0.34 (0.10)	0.41 (0.08)	0.36 (0.08)	0.33 (0.09)	0.31 (0.10)
Protected Area (% cover)	2.9 (6.0)	3.7 (8.7)	2.9 (5.9)	2.6 (4.7)	3.1 (6.3)
Population per Kilometer	2,989 (4,035)	2,150 (2,884)	2,964 (3,238)	3,250 (3,634)	2,998 (5,476)
¹ Mean (SD)					

Supplementary Table 2: Mixed effects models predicting whether or not a HOLC neighborhood sampled (left), sampling density (center), and sampling completeness (right) incorporating climatic information.

<i>Predictors</i>	log(sampling density)			log(sampling density)		
	<i>Estimates</i>	<i>CI</i>	<i>p</i>	<i>Estimates</i>	<i>CI</i>	<i>p</i>
(Intercept)	239.57	202.27 – 283.73	<0.001	175.60	10.71 – 2880.31	<0.001
holc grade [B]	0.63	0.54 – 0.73	<0.001	0.67	0.58 – 0.77	<0.001
holc grade [C]	0.37	0.31 – 0.43	<0.001	0.38	0.31 – 0.47	<0.001
holc grade [D]	0.26	0.21 – 0.31	<0.001	0.27	0.22 – 0.34	<0.001
ndvi	0.93	0.88 – 0.99	0.024	0.97	0.91 – 1.03	0.336
pct pa	1.45	1.39 – 1.51	<0.001	1.43	1.37 – 1.49	<0.001
pop per km	1.33	1.26 – 1.40	<0.001	1.30	1.24 – 1.37	<0.001
Random Effects						
σ^2	3.73			3.59		
τ_{00}	0.41	msa_NAME		3.63	msa_NAME	
τ_{11}	0.10	msa_NAME.holc_gradeB		0.00	msa_NAME.holc_gradeB	
	0.14	msa_NAME.holc_gradeC		0.62	msa_NAME.holc_gradeC	
	0.30	msa_NAME.holc_gradeD		0.46	msa_NAME.holc_gradeD	
				3.55	msa_NAME.mean_temp_c	
				4.53	msa_NAME.mean_precip_mm	
				0.00	msa_NAME.msa_medhhincE	
				0.01	msa_NAME.mean_temp_c:mean_precip_mm	
ϱ_{01}	1.00			-0.62		
	0.61			-0.00		
	0.12			-0.01		
				0.03		
				0.01		
				0.13		
				-0.33		
ICC				0.52		
N	143	msa_NAME		143	msa_NAME	
Observations	8905			8905		
Marginal R ² / Conditional R ²	0.098 / NA			0.047 / 0.547		
AIC	122260.698			124074.442		
AICc	122260.775			124074.889		

Supplementary Table 3: Mixed effect models using (I) Type I sum of squares and (II) Type II sum of squares.

(I) Sample binary

Model Summary				
Parameter	Sum_Squares	df	Mean_Square	F
holc_grade	25.18	3	8.39	8.39
Anova Table (Type 1 tests)				

Sampling density

Model Summary				
Parameter	Sum_Squares	df	Mean_Square	F
scale(ndvi)	6.59	1	6.59	1.77
scale(pct_pa)	1155.27	1	1155.27	309.48
scale(pop_per_km)	578.94	1	578.94	155.09
holc_grade	796.61	3	265.54	71.13
Anova Table (Type 1 tests)				

Completeness

Model Summary				
Parameter	Sum_Squares	df	Mean_Square	F
scale(ndvi)	12177.52	1	12177.52	25.43
scale(pct_pa)	42891.83	1	42891.83	89.59
scale(pop_per_km)	31222.24	1	31222.24	65.21
holc_grade	28979.22	3	9659.74	20.18
Anova Table (Type 1 tests)				

(II) Sample binary

Model Summary		
Parameter	Chi2(3)	p
holc_grade	25.68	< .001
Anova Table (Type 2 tests)		

Sampling density

Model Summary			
Parameter	Chi2	df	p
scale(ndvi)	5.07	1	0.024
scale(pct_pa)	304.91	1	< .001
scale(pop_per_km)	110.96	1	< .001
holc_grade	213.40	3	< .001
Anova Table (Type 2 tests)			

Completeness

Model Summary			
Parameter	Chi2	df	p
scale(ndvi)	8.30	1	0.004
scale(pct_pa)	94.72	1	< .001
scale(pop_per_km)	45.57	1	< .001
holc_grade	60.53	3	< .001
Anova Table (Type 2 tests)			

Supplementary Table 4: Generalized additive model predicting log number of bird biodiversity observations over time across HOLC grade (years 2000 to 2020). Biodiversity records are significantly different across HOLC grades over time with highest records numbers in HOLC A and lowest on HOLC D neighborhoods.

Component	Term	Estimate	Std Error	t-value	p-value	
A. parametric coefficients	(Intercept)	-400.140	7.094	-56.408	0.0000	***
	Year	0.202	0.004	57.122	0.0000	***
	holc_gradeB	138.364	10.032	13.792	0.0000	***
	holc_gradeC	67.074	10.032	6.686	0.0000	***
	holc_gradeD	73.204	10.032	7.297	0.0000	***
	Year:holc_gradeB	-0.069	0.005	-13.756	0.0000	***
	Year:holc_gradeC	-0.034	0.005	-6.713	0.0000	***
	Year:holc_gradeD	-0.037	0.005	-7.341	0.0000	***

Signif. codes: 0 '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.992, Deviance explained 0.992

GCV : 0.0106, Scale est: 0.00959, N: 84